Data Story 5: Quantifying Factors that Control Housing Sales Prices

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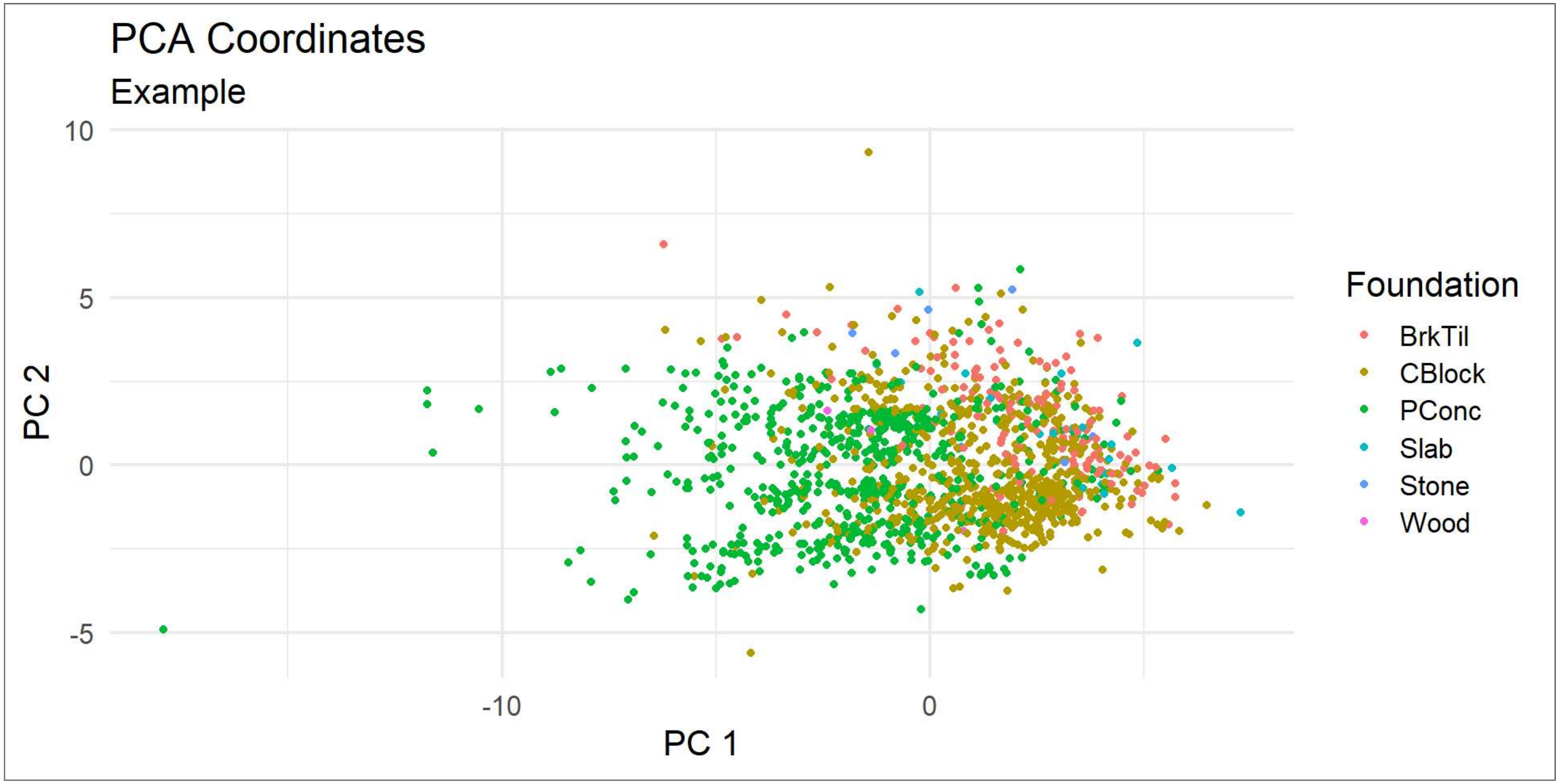
Quantifying Factors that Control Housing Sales Prices

Ames Dataset

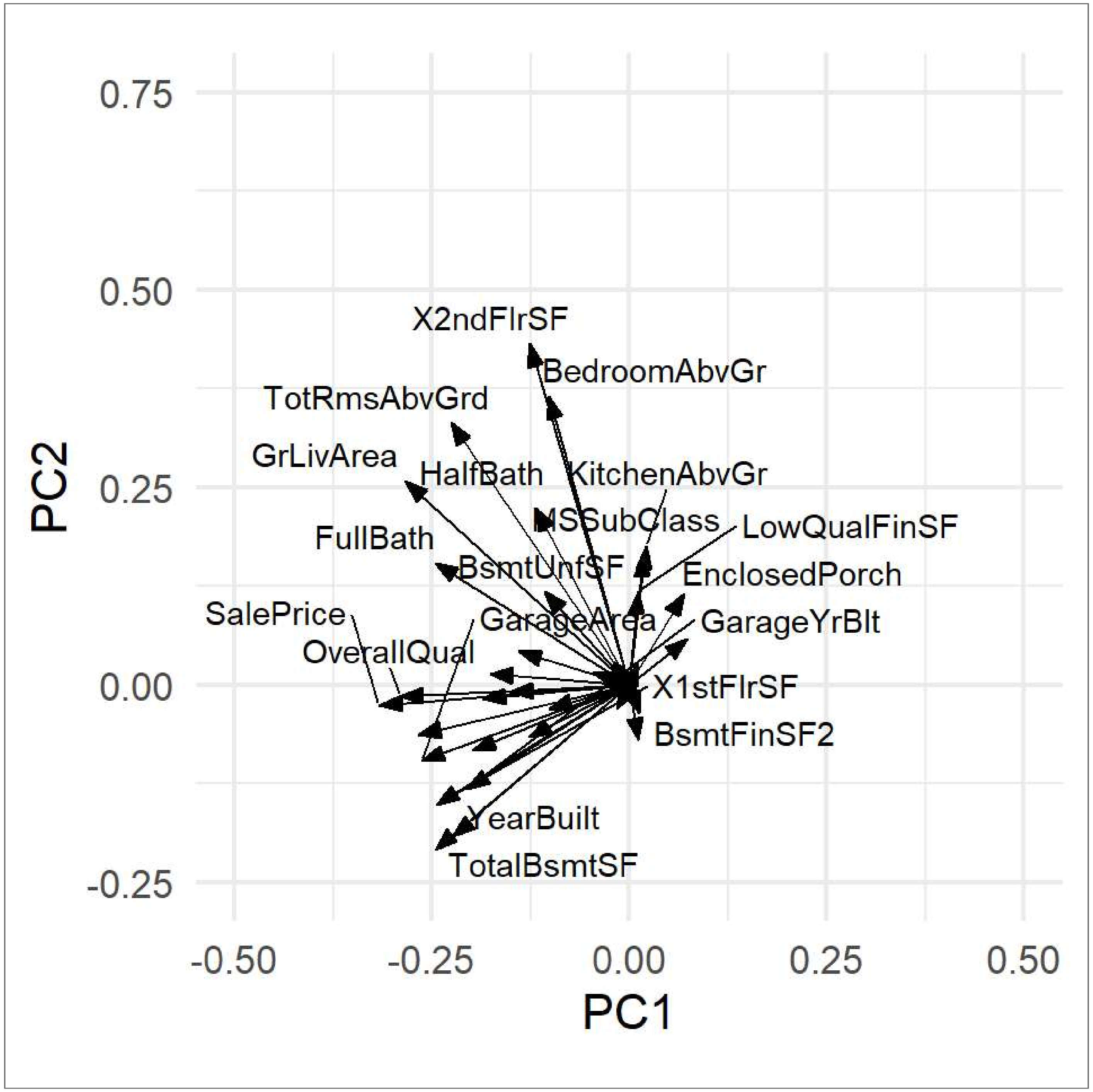
```
  Id MSSubClass MSZoning LotFrontage LotArea Street Alley LotShape LandContour
1  1         60       RL          65    8450   Pave  <NA>      Reg         Lvl
2  2         20       RL          80    9600   Pave  <NA>      Reg         Lvl
3  3         60       RL          68   11250   Pave  <NA>      IR1         Lvl
4  4         70       RL          60    9550   Pave  <NA>      IR1         Lvl
5  5         60       RL          84   14260   Pave  <NA>      IR1         Lvl
6  6         50       RL          85   14115   Pave  <NA>      IR1         Lvl
  Utilities LotConfig LandSlope Neighborhood Condition1 Condition2 BldgType
1    AllPub    Inside       Gtl      CollgCr       Norm       Norm     1Fam
2    AllPub       FR2       Gtl      Veenker      Feedr       Norm     1Fam
3    AllPub    Inside       Gtl      CollgCr       Norm       Norm     1Fam
4    AllPub    Corner       Gtl      Crawfor       Norm       Norm     1Fam
5    AllPub       FR2       Gtl      NoRidge       Norm       Norm     1Fam
6    AllPub    Inside       Gtl      Mitchel       Norm       Norm     1Fam
  HouseStyle OverallQual OverallCond YearBuilt YearRemodAdd RoofStyle RoofMatl
1     2Story           7           5      2003         2003     Gable  CompShg
2     1Story           6           8      1976         1976     Gable  CompShg
3     2Story           7           5      2001         2002     Gable  CompShg
4     2Story           7           5      1915         1970     Gable  CompShg
5     2Story           8           5      2000         2000     Gable  CompShg
6     1.5Fin           5           5      1993         1995     Gable  CompShg
  Exterior1st Exterior2nd MasVnrType MasVnrArea ExterQual ExterCond Foundation
```

Principal Component Analysis On Ames Dataset Numeric Values

Dataset can be dimensionally reduced PC1 and PC2 graphed below

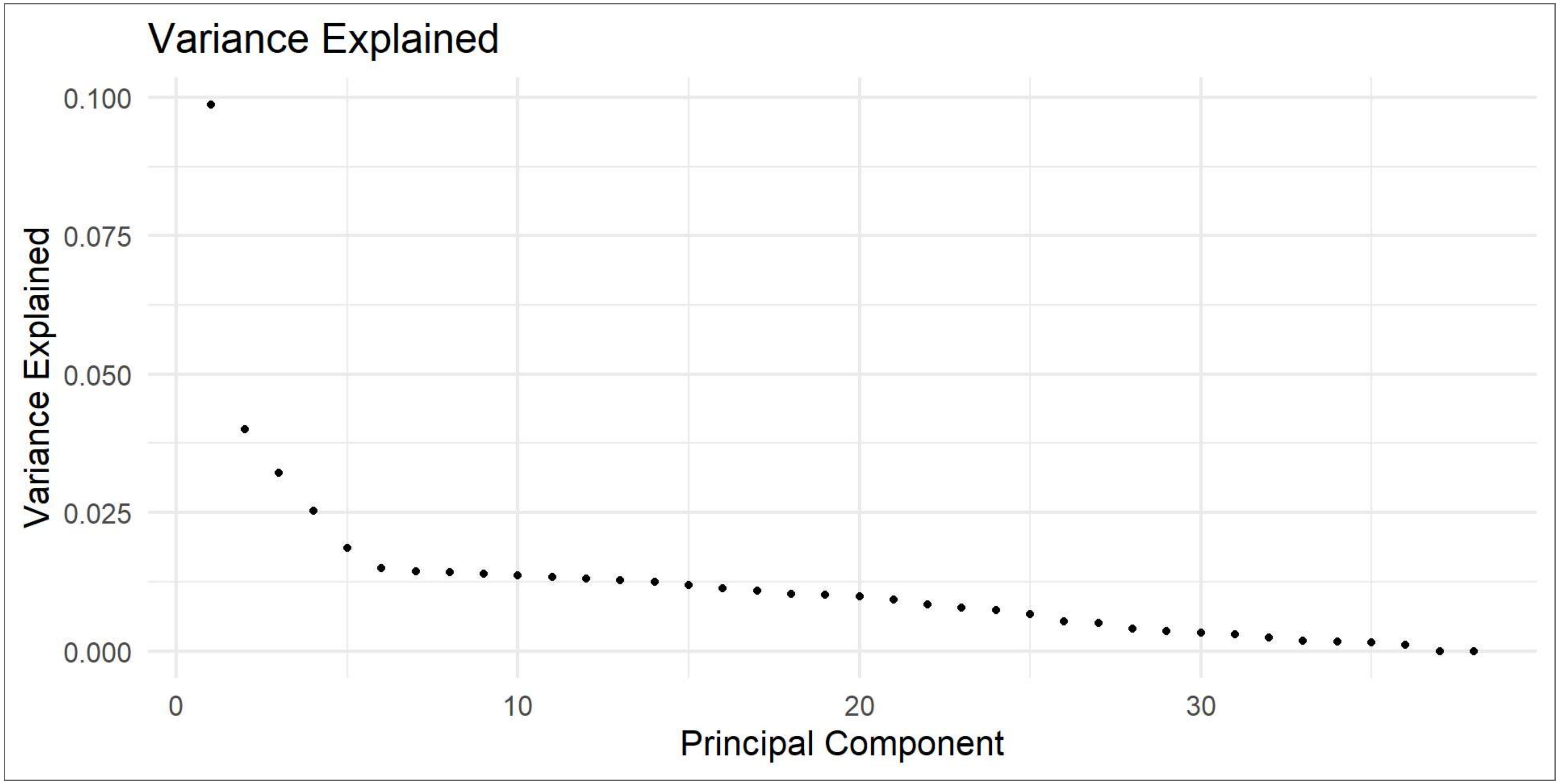


What Variables Are the Principal Components?



How Many Principle Components should we use?

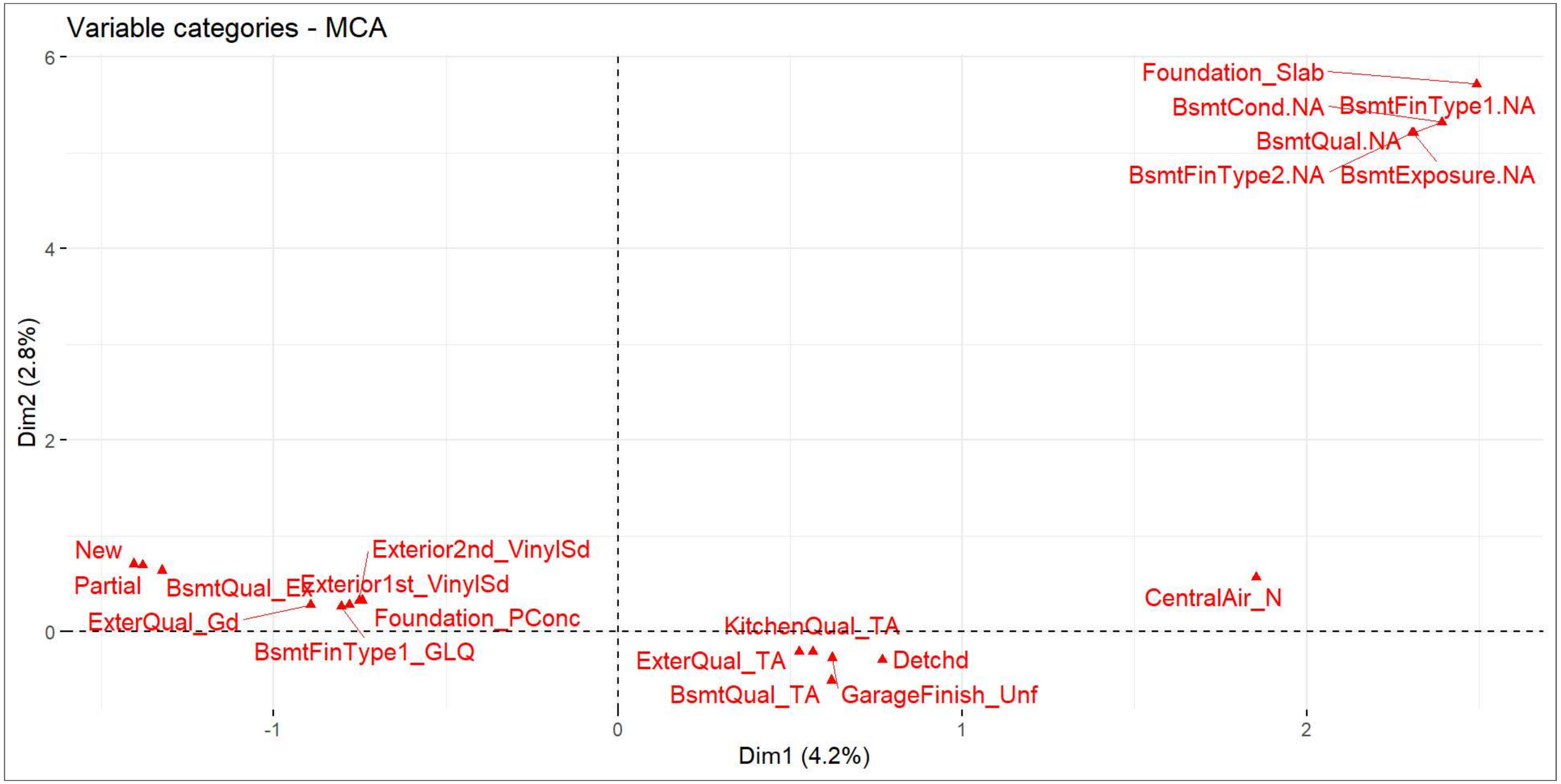
~5



Variables within Principal Components 1-5

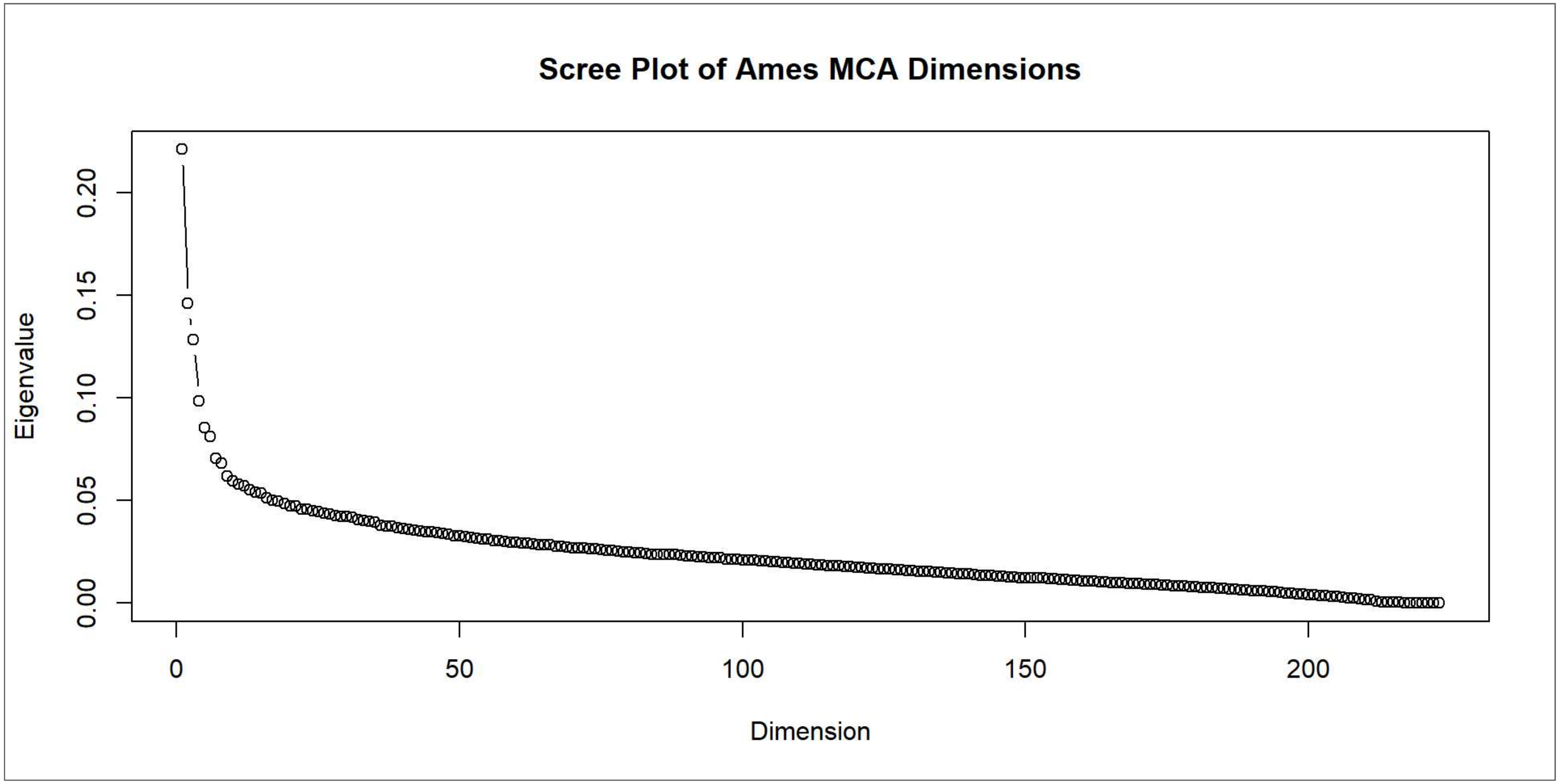
```
      variable         PC1        PC2         PC3         PC4         PC5
1 KitchenAbvGr 0.000452842 0.03066053 0.003239022 0.002353456 0.332828260
2    BsmtUnfSF 0.011596592 0.01382149 0.011284776 0.319454425 0.003668325
3 BsmtFullBath 0.005495671 0.08340290 0.024265028 0.124503630 0.050499620
4   BsmtFinSF1 0.018768304 0.08377171 0.047529223 0.109552621 0.024084150
5    X2ndFlrSF 0.016053065 0.18645880 0.004618138 0.068051975 0.007622644
6   MSSubClass 0.000321077 0.02740589 0.050445062 0.088799119 0.088807802
  total_contrib
1     0.3695341
2     0.3598256
3     0.2881668
4     0.2837060
5     0.2828046
6     0.2557790
```

Multi Component Analysis on Categorical Variables



How many MCA components to use?

again ~5



Which categorical variables to use?

```
# A tibble: 10 × 2
   variable     total_contrib
   <chr>                <dbl>
 1 Foundation            6.39
 2 ExterQual             5.96
 3 BsmtQual              5.46
 4 KitchenQual           5.33
 5 Exterior1st           4.69
 6 Exterior2nd           4.65
 7 GarageFinish          4.31
 8 HeatingQC             3.87
 9 BsmtFinType1          2.78
10 MasVnrType            2.62
```

```
# A tibble: 10 × 2
   variable        total_contrib
   <chr>                   <dbl>
 1 BsmtCond.NA             11.7
 2 BsmtFinType1.NA         11.7
 3 BsmtQual.NA             11.7
 4 BsmtFinType2.NA         11.5
 5 BsmtExposure.NA         11.5
 6 Foundation              10.9
 7 BsmtQual                 2.70
 8 Exterior1st              2.38
 9 Wall                     2.05
10 Exterior2nd              2.03
```

```
# A tibble: 10 × 2
   variable        total_contrib
   <chr>                   <dbl>
 1 Exterior1st              7.80
 2 Exterior2nd              7.57
 3 Foundation               6.78
 4 BsmtFinType1             4.13
 5 GarageCond.NA            3.09
 6 GarageFinish.NA          3.09
 7 GarageQual.NA            3.09
 8 GarageType.NA            3.09
 9 KitchenQual              2.94
10 ExterQual                2.58
```

```
# A tibble: 10 × 2
   variable        total_contrib
   <chr>                   <dbl>
 1 GarageCond.NA           11.9
 2 GarageFinish.NA         11.9
 3 GarageQual.NA           11.9
 4 GarageType.NA           11.9
 5 Exterior2nd              4.44
 6 Detchd                   3.98
 7 Exterior1st              3.88
 8 Foundation               3.34
```

```
 9 GarageFinish             2.39
10 GarageQual               2.35
```

```
# A tibble: 10 × 2
   variable     total_contrib
   <chr>                <dbl>
 1 RoofStyle             8.92
 2 LandSlope             6.31
 3 ExterQual             5.60
 4 Exterior1st           5.45
 5 BsmtExposure          5.38
 6 KitchenQual           5.32
 7 BsmtQual              5.26
 8 Exterior2nd           4.73
 9 Tar&Grv               4.40
10 Low                   3.48
```

Building a Model

Using variables from the top 5 PCA and MCA dimensions

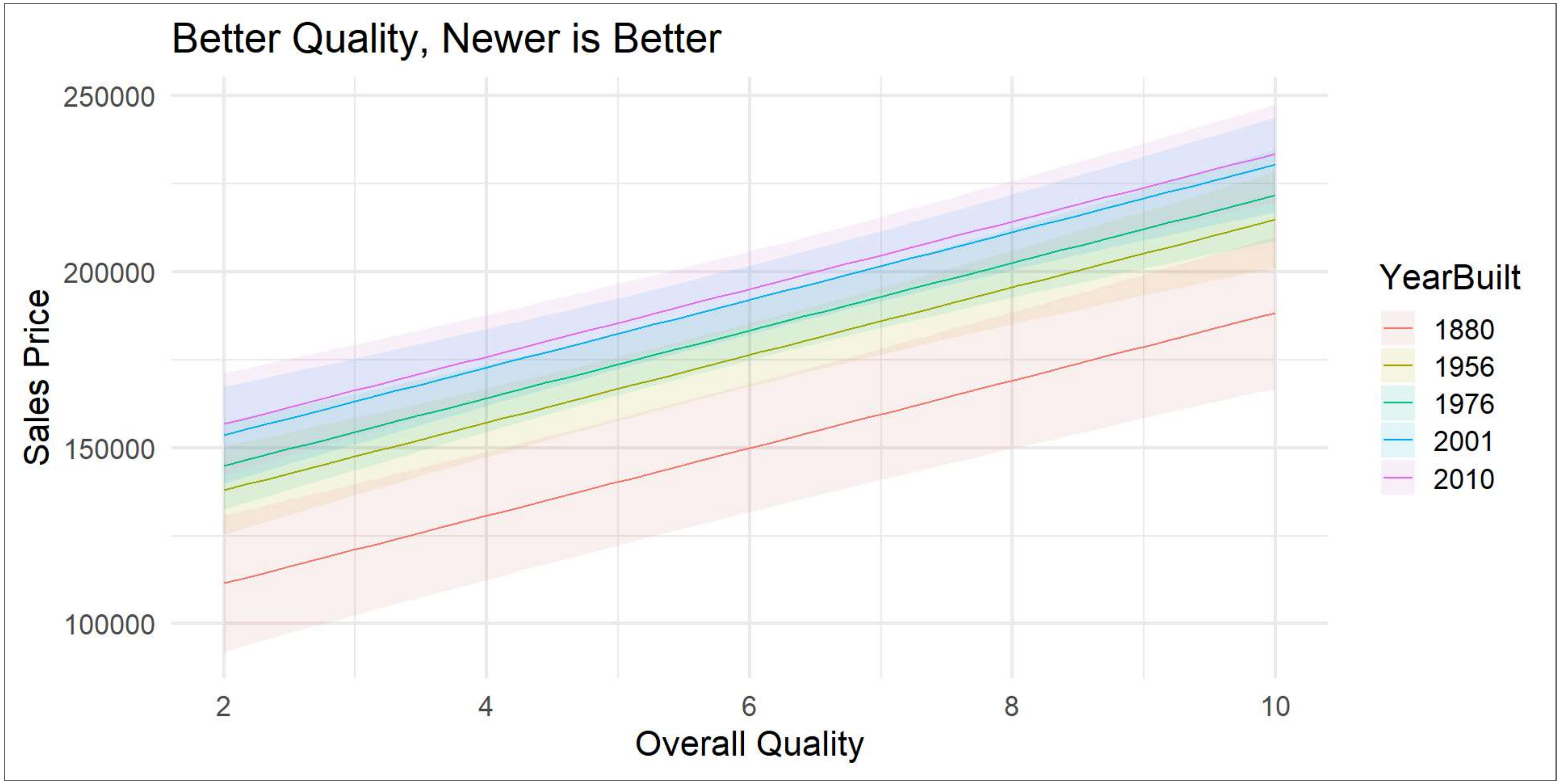
```
Call:
lm(formula = SalePrice ~ Foundation + ExterQual + BsmtQual +
    KitchenQual + Exterior1st + Exterior2nd + GarageFinish +
    HeatingQC + BsmtFinType1 + MasVnrType + BsmtCond + BsmtFinType2 +
    BsmtExposure + GarageCond + GarageFinish + GarageQual + GarageType +
    KitchenQual + ExterQual + RoofStyle + LandSlope + RoofMatl +
    OverallQual + GrLivArea + GarageCars + GarageArea + TotalBsmtSF +
    FullBath + X1stFlrSF + TotRmsAbvGrd + YearBuilt + X2ndFlrSF +
    BsmtFinSF1 + BsmtFullBath + HalfBath + KitchenAbvGr + GarageYrBlt +
    LotArea + LotFrontage + YearRemodAdd + MSSubClass + Fireplaces +
    EnclosedPorch + BsmtUnfSF + WoodDeckSF + OverallCond + BsmtHalfBath +
    ScreenPorch, data = train_df)

Coefficients:
      (Intercept)    FoundationCBlock     FoundationPConc     FoundationStone
       -1.427e+06           2.330e+03           7.359e+03           4.911e+03
   FoundationWood         ExterQualFa         ExterQualGd         ExterQualTA
       -2.695e+04          -3.570e+04          -2.080e+04          -2.684e+04
       BsmtQualFa          BsmtQualGd          BsmtQualTA       KitchenQualFa
       -2.113e+04          -2.591e+04          -2.607e+04          -2.347e+04
    KitchenQualGd       KitchenQualTA  Exterior1stBrkComm  Exterior1stBrkFace
```

Components of the Model sorted by p values

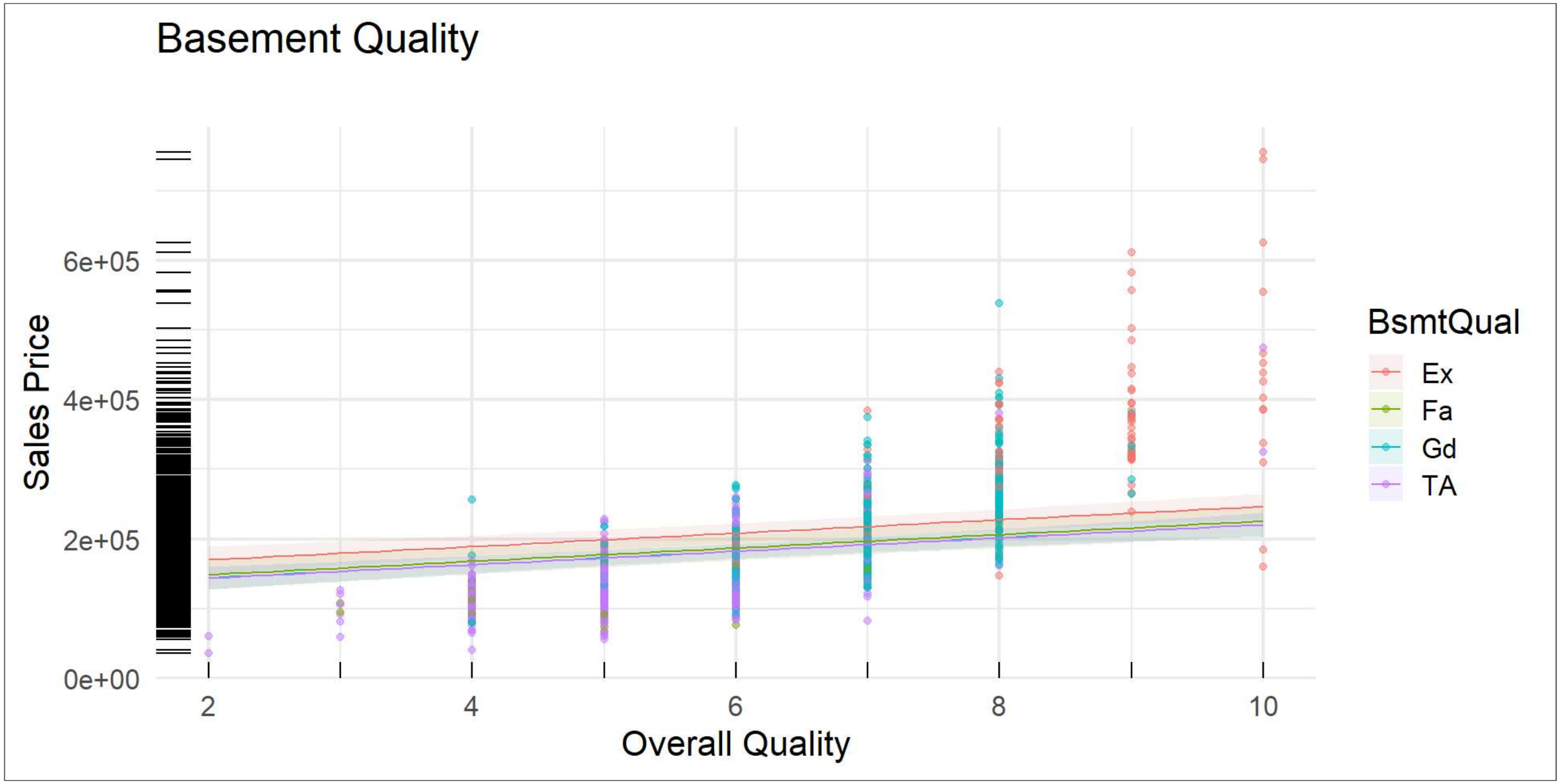
	Estimate	Std. Error	t value	Pr(>|t|)
RoofMatlWdShngl	7.231745e+05	3.918030e+04	18.457605	2.788635e-67
RoofMatlCompShg	6.739376e+05	3.758531e+04	17.930881	5.355702e-64
RoofMatlWdShake	6.757257e+05	4.172293e+04	16.195550	1.387784e-53
RoofMatlTar&Grv	6.757834e+05	4.356694e+04	15.511379	1.157149e-49
RoofMatlRoll	6.744639e+05	4.855997e+04	13.889300	7.906505e-41
RoofMatlMetal	7.467218e+05	5.386402e+04	13.863090	1.084186e-40
RoofMatlMembran	7.530146e+05	5.440896e+04	13.839899	1.433095e-40
OverallQual	9.614429e+03	1.209081e+03	7.951850	4.160092e-15
BsmtQualGd	-2.591202e+04	3.809124e+03	-6.802618	1.605521e-11
(Intercept)	-1.426570e+06	2.128468e+05	-6.702335	3.126577e-11
KitchenQualGd	-2.676296e+04	4.189370e+03	-6.388301	2.380671e-10
OverallCond	6.310216e+03	1.050372e+03	6.007601	2.481807e-09
KitchenQualTA	-2.696006e+04	4.747887e+03	-5.678328	1.698681e-08
BsmtQualTA	-2.607118e+04	4.792464e+03	-5.440037	6.431576e-08
LotArea	6.108776e-01	1.142715e-01	5.345843	1.073381e-07
KitchenAbvGr	-2.581576e+04	5.875442e+03	-4.393842	1.210371e-05
ExterQualTA	-2.684462e+04	6.207008e+03	-4.324889	1.650604e-05
YearBuilt	3.479200e+02	8.838291e+01	3.936508	8.734804e-05
ExterQualGd	-2.079855e+04	5.567961e+03	-3.735399	1.960742e-04
MSSubClass	-9.819464e+01	2.670023e+01	-3.677670	2.456171e-04

term

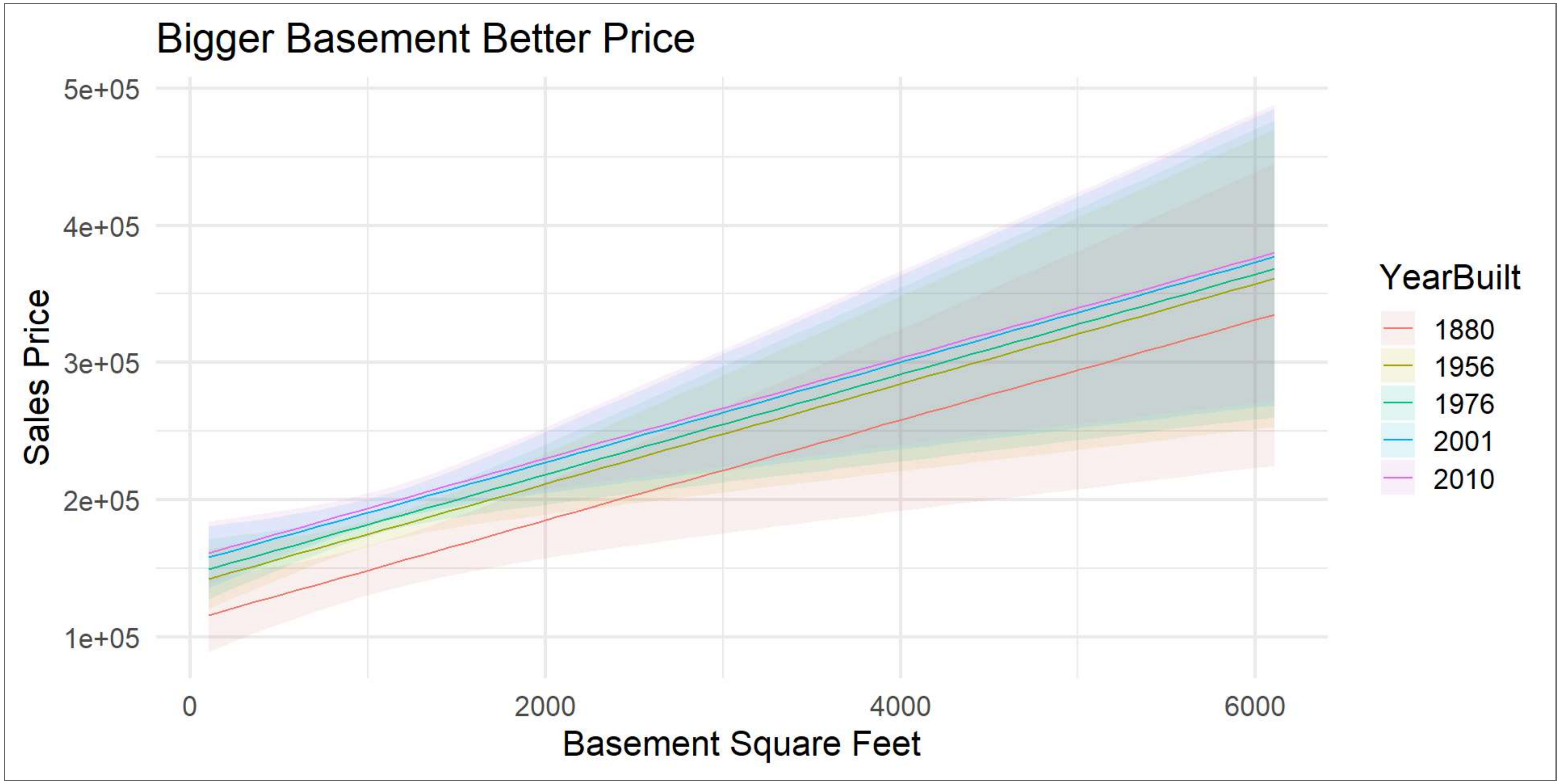
Model Sales Price Prediction. Overall Quality and How New?



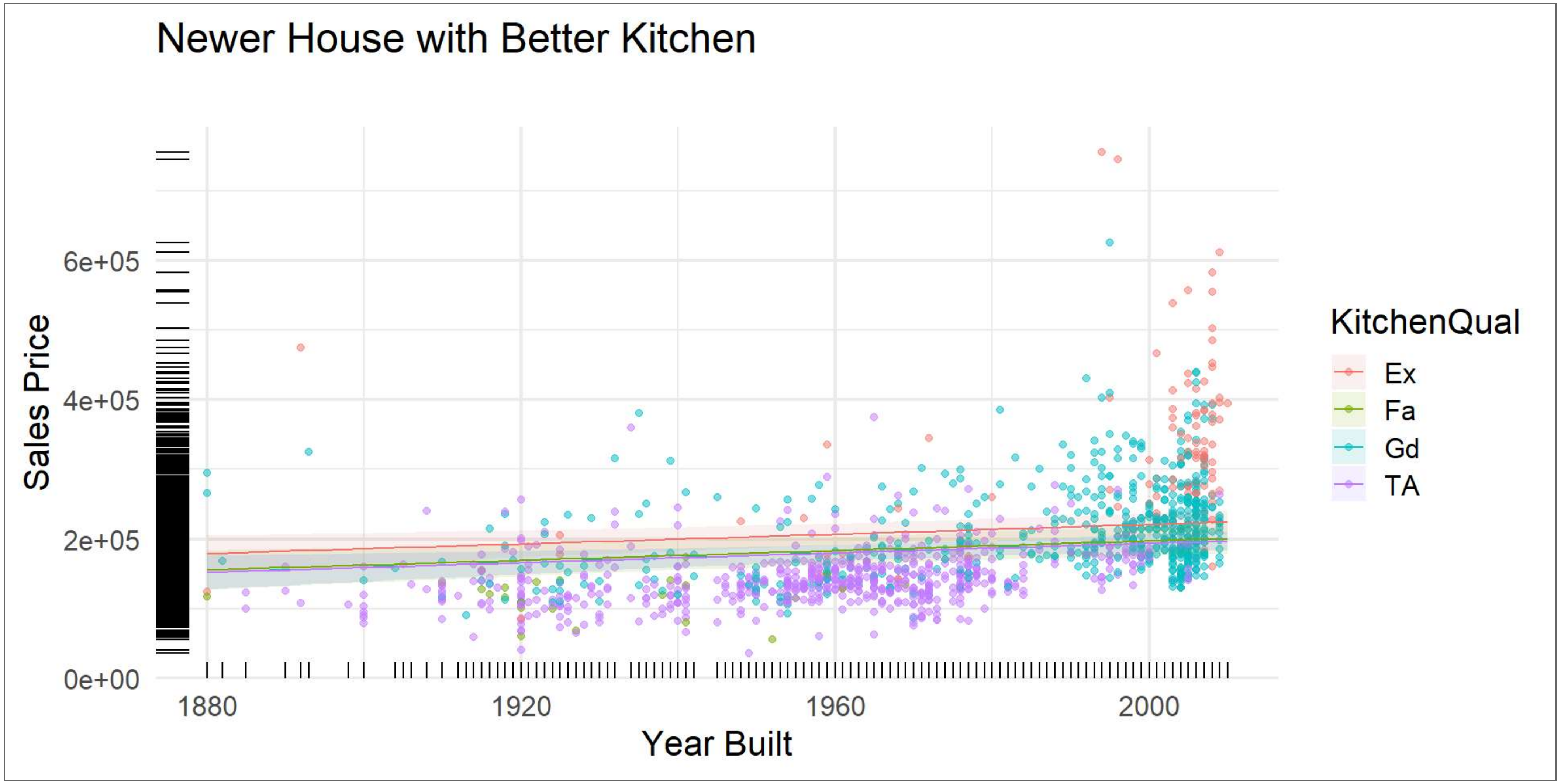
A Better Basement Makes a Difference



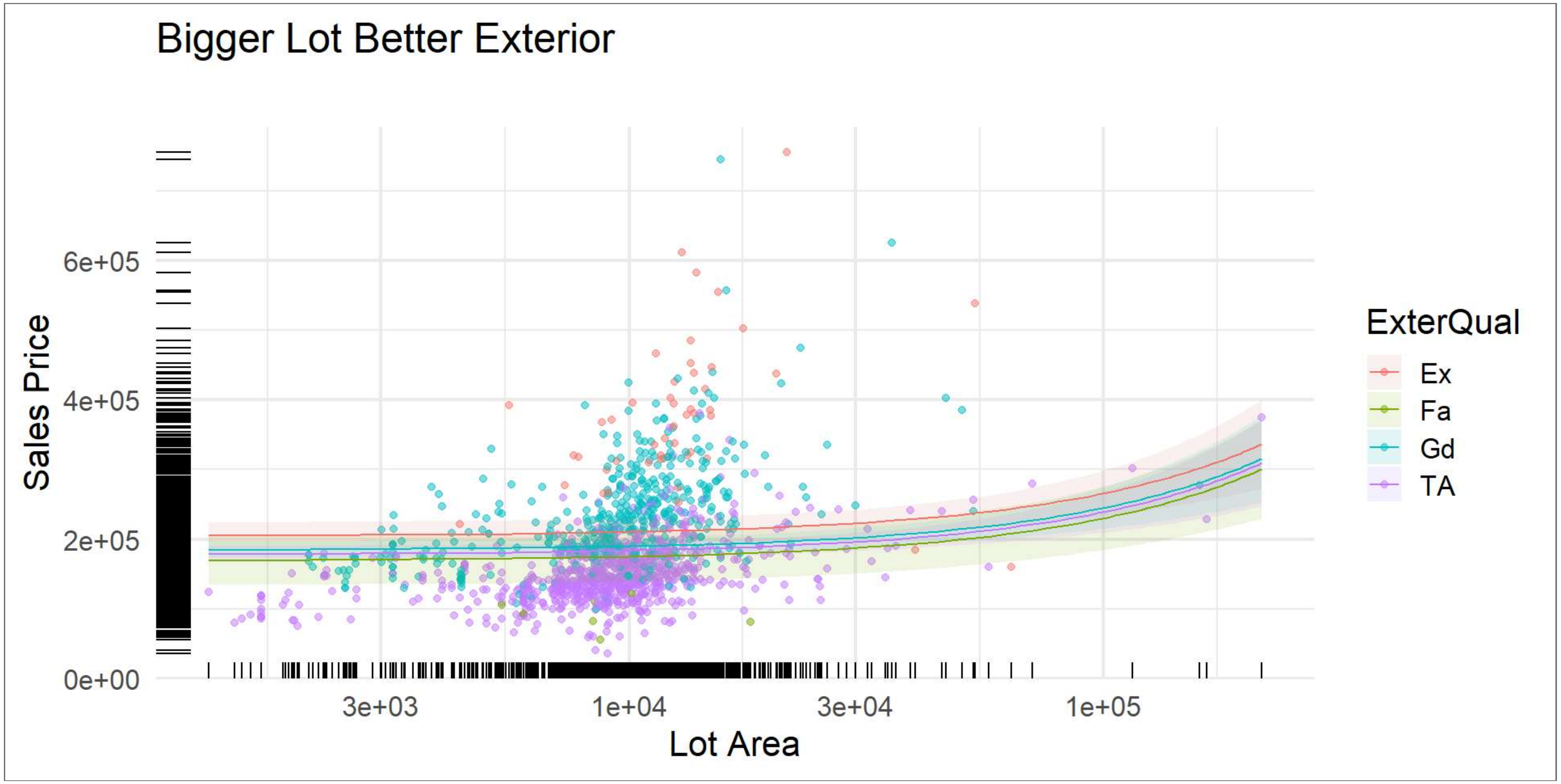
Size of Basement Matters too



So Does the Kitchen



Lot Size and Exterior Quality



Speaker notes